Review Problems
AP Calculus
Sec 6.1 – 6.9 Name_____

These are sample questions from unit 6. This is not an exact review for the test, just practice problems.

Multiple Choice:

1. Let $f(x)$ be a continuous function. What is the value of $\int_3^{15} f(x)\,dx$ if it is known that $\int_1^5 f(3x)\,dx = 7$?

(A) $\frac{7}{3}$ (B) 7

(C) 21 (D) $\frac{3}{7}$

2. Choose the correct statement(s) given that $\int_0^9 f(x)\,dx = 11$ and $\int_3^9 f(x)\,dx = -1$.

I. $\int_9^3 f(x)\,dx = 1$ **II.** $\int_3^0 f(x)\,dx = 6$ **III.** $\int_0^3 f(x)\,dx = 12$

(A) I only (B) II only (C) III only (D) I and II (E) I and III

3. If the substitution $u = x - 1$ is made, then the integral $\int_2^5 x\sqrt{x-1}\ dx =$

(A) $\int_2^5 (u+1)\sqrt{u}\ du$ (B) $\int_1^4 (u+1)\sqrt{u}\ du$

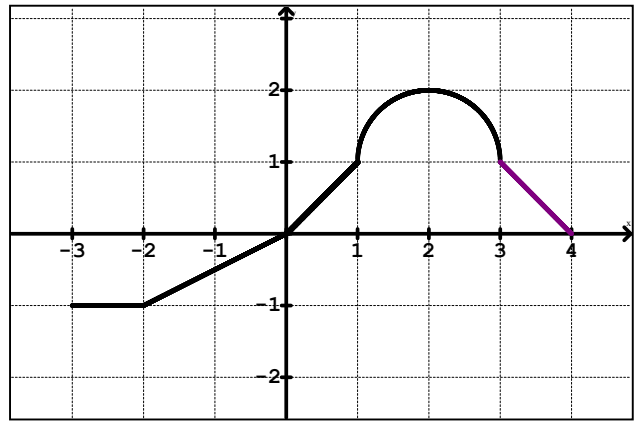
(C) $\int_1^4 u\sqrt{u-1}\ du$ (D) $\int_2^5 u\sqrt{u-1}\ du$

4. $\lim_{n\to\infty} \sum_{k=1}^{n} \left(\sin^2\left(1 + \frac{4k}{n}\right) + 3\right)\left(\frac{4}{n}\right) =$

(A) $\int_0^4 (\sin^2 x + 3)dx$ (B) $\int_1^5 (\sin^2 x + 3)dx$

(C) $4\int_0^1 (\sin^2(1+x) + 3)dx$ (D) $\int_0^1 (\sin^2(1+4x) + 3)dx$

Use the graph of $f(x)$ below for problems **5** and **6.**



5. Find $\int_1^4 f(x)\,dx$.

(A) $\frac{\pi+5}{2}$ (B) $\frac{5}{2}+\pi$

(C) $\pi+3$ (D) $\frac{\pi}{2}+3$

6. Find $\int_{-3}^1 f(x)\,dx$.

(A) -2.5 (B) -1.5

(C) 3 (D) 2.5

7. If the substitution $u = \cos 3x$ is made, then the integral $\int_0^{\frac{\pi}{6}} \cos^3 3x \sin 3x \; dx =$

(A) $-\frac{1}{3}\int_0^{\frac{\pi}{6}} u^3\,du$ (B) $\frac{1}{3}\int_0^{\pi} u^3\,du$

(C) $-\frac{1}{3}\int_1^{0} u^3\,du$ (D) $-\frac{1}{3}\int_0^{1} u^3\,du$

8. Consider $F(x) = \int_1^{2x} \frac{1}{1+t^3}\,dt$. Find $F'(x)$.

(A) $\frac{1}{1+8x^2}$ (B) $\frac{1}{1+2x^2}$

(C) $\frac{2}{1+8x^3}-\frac{1}{2}$ (D) $\frac{2}{1+8x^3}$

Free Response Questions

1. Evaluate the integral.

$$\int \frac{16x^4 + 8x^2 + 3x}{x} \mathrm{d}x$$

2. Evaluate the integral.

$$\int \left(30x^4 - \frac{14}{x^3}\right) \mathrm{d}x$$

3. Calculate the integral and write your answer in simplest form.

$$\int \frac{3}{\sqrt{x^5}} \mathrm{dx}$$

4. Calculate the integral and write your answer in simplest form.

$$\int \frac{3\sqrt[3]{x^5}}{2} \mathrm{dx}$$

5. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int 4x^2\sqrt[3]{4x^3 + 5}\, \mathrm{d}x$$

6. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int 3x(1 - 3x^2)^{\frac{3}{2}}\, \mathrm{d}x$$

7. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int \frac{2}{3\sqrt[3]{6x + 1}}\, \mathrm{d}x$$

8. Use u-substitution to evaluate the integral and express your answer in simplest form.

$$\int \frac{18x}{(x^2 + 3)^4}\, \mathrm{d}x$$

9. Calculate the integral and express your answer in simplest form.

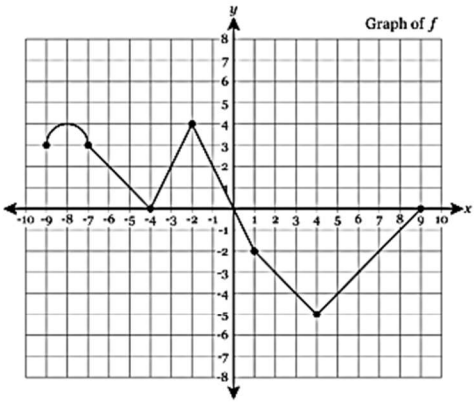
$$\int_0^1 \frac{3x^2\sqrt{9 - 5x^3}}{2}\, \mathrm{d}x$$

10. Calculate the integral and express your answer in simplest form.

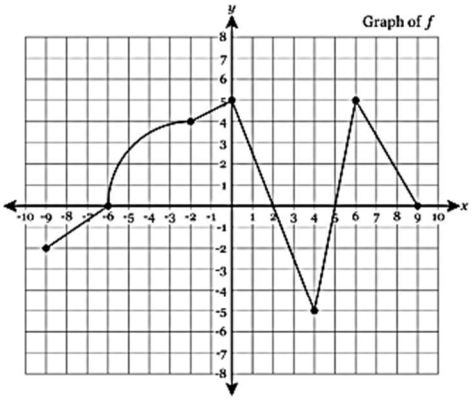
$$\int_{-1}^1 \frac{9x^2}{(5 - x^3)^2}\, \mathrm{d}x$$

11. Evaluate $\frac{d}{dx}\int_3^{5x} \cos(t^2)\, dt$

12. The graph of the function f in the figure below consists of line segments and a semicircle. Let g be the function given by $g(x) = \int_{-4}^{x} f(t)dt$. Determine all intervals on the closed interval $[-9, 9]$ where g is concave up.



13. The graph of the function f in the figure below consists of line segments and a quarter of a circle. Let g be the function given by $g(x) = \int_{5}^{x} f(t)dt$. Determine all values of x, if any, where g has a relative maximum on the open interval $(-9, 9)$.



14. Given the function $f(x) = \int_{1}^{x} \left(t^3 - 18t^2 + 81t\right) dt$, determine all x-values, if any, for which f has a point of inflection.

15. Evaluate the integral and express your answer in simplest form.

$$\int \frac{2}{\sqrt{25 - 16x^2}} dx$$

16. Evaluate the integral and express your answer in simplest form.

$$\int \frac{-8}{1 + 16x^2} dx$$

17. Let f be a continuous function such that $\int_{2}^{10} f(x)\, dx = -3$ and $\int_{-1}^{10} f(x)\, dx = 3$. What is the value of $\int_{-1}^{2} \left(f(x) + 1\right) dx$?

18. Let f be a continuous function such that $\int_{1}^{8} f(x)\, dx = 2$ and $\int_{3}^{1} f(x)\, dx = 3$. What is the value of $\int_{3}^{8} \left(f(x) + 5\right) dx$?